

Dual-Regime Insert/Abort: Sensor Perturbation vs. Selection Shift

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Abstract

Temperature scaling and related post-hoc maps are often treated as a cheap fix for miscalibrated probabilities. We measure whether that fix transfers under two different test-time changes, for logistic regression, random forests, and histogram gradient boosting on small tabular datasets (no deep nets, no API models). *Feature mean-shift with frozen labels*—a perturbation of X that does **not** preserve $P(Y | X)$ —raises ECE: at strength 1.5, uncalibrated $\Delta\text{ECE} = \text{ECE}_{\text{pert}} - \text{ECE}_{\text{iid}}$ is positive in 51/60 seed-level cells (mean 0.113, cell-to-cell sd 0.149). That pool mixes sklearn sets (mean 0.042, 25/30 positive) with synthetics (mean 0.184, 26/30 positive) where accuracy also falls. Averaging seeds first, all 12/12 uncalibrated dataset $\times$ model means are positive (a sign count; models that share a dataset are not independent). After i.i.d. temperature, isotonic, or histogram maps, pooled ΔECE remains positive. The same maps can help or hurt relative to doing nothing. By contrast, *selection* on X that keeps the original (X, y) pairs—closer to covariate shift, but a milder change in accuracy—yields smaller ΔECE (quantile-slice cell-mean 0.016 on 6 dataset $\times$ model cells from 2 datasets). Unweighted split-conformal coverage falls under feature perturbation. On a planar peg-in-hole *insert vs. abort* cartoon (geometric clearance, not a robot), an optimistic encoder—frozen labels, reports the peg closer to the hole than the true pose—makes source temperature on *raw* encoder probabilities a false-confident insert policy (FCAR 0.342, mean utility -3.000). A physics-residual channel that *projects* the encoder onto the camera pose and only then applies the same source T recovers utility 0.093 (FCAR 0.014, 5 seeds) and beats abort-all (-0.500). A pairing-preserving fixture slice is difficulty-matched (success rate 0.426 vs. i.i.d. 0.458) and is not aborted as a sensor fault (abort rate 0.000). This is an analysis-plus-system result, not a new calibrator and not hardware.

1 Introduction

A classifier is *calibrated* when a predicted probability p is correct with frequency p . Post-hoc maps—Platt-style scaling, histogram binning, isotonic regression, and temperature scaling—are the standard way to improve calibration without retraining [1–3]. A separate literature shows that deep-net un-

certainty *collapses* under dataset shift [4–6]. Almost all of that evidence is vision-scale neural nets. Tree ensembles remain strong on tabular data [7], and their i.i.d. calibration properties have been known for decades [2]. Within our API-verified bibliography—mostly deep-net calibration and conformal theory, not a systematic search of applied tabular-drift venues—we did not find this exact measurement (sklearn models, i.i.d. post-hoc maps, test-time feature perturbation vs. selection). That is a scoped gap, not a proof of absence.

We treat that as a measurement question. We do not introduce a new calibrator. A negative or mixed answer is an acceptable outcome.

Primary hypothesis (reframed). I.i.d.-fitted post-hoc maps do not keep ECE from rising under a Gaussian mean-shift of test features with *labels held fixed* (input perturbation / a change in $P(Y | X)$ at the observed x). We do **not** claim that genuine covariate shift—selection on X that preserves pairings—has the same effect; that is tested as a control.

Two other hypotheses were piloted and *not* selected as the paper claim: isotonic overfit vs. temperature as a function of n_{cal} (H2; heterogeneous), and oracle-weighted conformal recovery (H3; the 1-D weight estimator failed). Decision log: logs/decisions.md.

2 Related work

Surveys of calibration methods and metrics include [8, 9]. ECE via confidence binning [3, 10] is biased and definition-dependent [11–13]; we therefore also report Brier score [14] and NLL [15] (pooled Δ alongside ECE in Results). Temperature scaling [3] and Dirichlet calibration [16] target neural nets. Unsupervised recalibration under covariate shift exists for neural models (Pampari and Ermon [17]). Domain-drift post-hoc calibration of CNNs is studied by Tomani et al. [6]. Those papers already *predict* our headline outcomes: ECE should rise under shift, and i.i.d.-fitted post-hoc maps should not be a complete fix. A surprising result here would have been that logistic regression, random forests, and histogram boosting *stay* calibrated under a mean-shift of used covariates *with labels frozen*. Selection that preserves (X, y) pairs is a different mechanism. We reuse the questions, not the model class.

Conformal prediction has finite-sample coverage under exchangeability [18–20] and requires density-ratio weights under covariate shift [21]. We use unweighted split conformal as a secondary diagnostic, not as a proposed fix. Conformal warning systems for grasping and driving [22] and perception-based safe control [23, 24] treat shift as one blob. Density-ratio conformal [21] is the legal act policy only while $P(Y | X)$ is invariant, which frozen-label encoder bias violates. We do not claim a method gap versus those papers. The system question here is whether unlabeled residual vs. density channels can route *opposite* legal moves on a DGP where those assumptions split.

3 Methods

Models. Scikit-learn logistic regression (with standard scaling), random forests, and HistGradientBoostingClassifier, with hyperparameters as in `src/calibshift/models.py`. All expose `predict_proba`.

Calibrators. Fitted on a held-out calibration split of the *source* distribution, then frozen. *none*: identity. *temperature*: one scalar T on $\log p$, fitted by NLL (Guo-style, using log-probabilities so trees without logits still apply). *isotonic*: sklearn isotonic regression on $p(y=1)$ (binary) or on confidence (multiclass). *histogram*: equal-width binning of the same 1-D score.

Shifts. *Feature perturbation*: add $s \cdot \hat{\sigma}_j$ to listed test columns, **keeping the original labels**. This moves x without moving y , so it is *not* covariate shift ($P(Y | X)$ is not invariant). $\hat{\sigma}$ is the test-split feature sd. *Selection*: quantile slice and importance resampling of existing (X, y) pairs (closer to covariate shift). Trailing-column perturbation is a control.

Metrics. Equal-width confidence ECE with 15 bins (from the exp04 config), Brier, NLL, accuracy, and split conformal coverage with the inverse-probability (LAC) score $1 - p_y$ at $\alpha = 0.1$ (not APS).

Protocol. Train/calibration/test partitions follow `calibshift.runner` (source-distribution splits; calibration data never includes the test set), stratified, with the 5 seeds listed in the exp04 config. Results are written only by `calibshift.io.write_result`. `paper/numbers.tex` is generated from those JSON files.

4 Experiments

Datasets. Offline-only: sklearn `breast_cancer` (binary) and `wine` (3-class); two `make_classification` sets (binary and 3-class). No OpenML, no paid APIs.

Table 1: Seed-pooled ΔECE at feature-perturbation $s=1.5$. “std” is heterogeneity, not se of the mean. The $n = 60$ rows are 4 datasets $\times$ 3 models $\times$ 5 seeds (pseudo-replicated). Sklearn/synthetic rows are uncalibrated subsets. One-sided Wilcoxon p -values on this pool are exploratory and live in `results/stats_tests.json` (22 tests, unadjusted).

Calibrator	mean ΔECE	std	#pos/neg
none	0.113	0.149	51/9
none (sklearn)	0.042	0.039	25/5
none (synthetic)	0.184	0.183	26/4
temperature	0.124	0.150	56/4
isotonic	0.141	0.151	53/7
histogram	0.141	0.151	57/3

Main grid (exp04 main_h1). 240 cells, 0 failed, 23.0 seconds on CPU. This is the source of all headline numbers.

Ablations. exp05: quantile slice and importance resampling. exp06: calibration-set size (killed primary H2). exp07: shift trailing columns.

Sanity. I.i.d. logistic regression on breast-cancer: accuracy 0.979, ECE 0.032 ± 0.010 (linear models on this set are typically $\gtrsim 0.95$ accurate). Wine random forests are overconfident i.i.d. (ECE 0.147 ± 0.020), matching the classical tree-calibration picture [2]. Split conformal coverage at $s=0$ is near the nominal $1 - \alpha = 0.9$ for $\alpha=0.1$. We do *not* compare ECE magnitudes to ImageNet temperature scaling [3].

5 Results

5.1 Feature perturbation usually raises ECE

Figure 1 plots shifted ECE against perturbation strength s . Figure 2 shows mean ΔECE at $s=1.5$. Pooled over 60 dataset $\times$ model $\times$ seed cells, uncalibrated mean $\Delta\text{ECE} = 0.113$ with cell-to-cell sd 0.149 (51 positive, 9 negative). The same cells have mean ΔBrier 0.257, mean ΔNLL 0.792, and mean accuracy change -0.156. Sklearn-only cells (breast-cancer and wine) have mean $\Delta\text{ECE} = 0.042$ (25/30 positive); synthetics have mean 0.184 (26/30 positive). After i.i.d. temperature scaling the pooled mean ΔECE is 0.124 (sd 0.150). These sd figures describe heterogeneity, not a confidence interval for the mean. Averaging seeds first, all 12 of 12 *uncalibrated* dataset $\times$ model means are positive. An exploratory one-sided Wilcoxon on those 12 means gives $p=2.4e-04$, but models that share a dataset are not independent; a dataset-level one-sided sign test on 4 dataset means is $p=0.0625$. Breast-cancer histogram boosting is the smallest cell (mean 0.002, seed sd 0.009). Synthetic logistic regression is the largest (mean 0.292, sd 0.214) and coincides with a large accuracy drop. Table 1 records seed-pooled sign counts, not confirmatory p -values; `results/stats_tests.json` contains 22 related one-sided tests with no multiplicity correction.

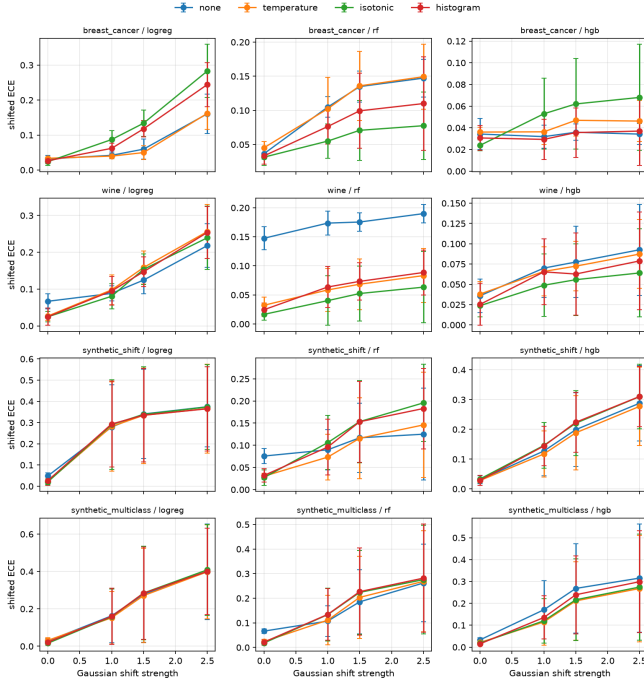


Figure 1: Shifted ECE vs. Gaussian feature-perturbation strength (mean $\pm$ seed sd). $s=0$ is i.i.d. Labels are frozen; this is not covariate shift. Source: results/exp04_main.h1.json.

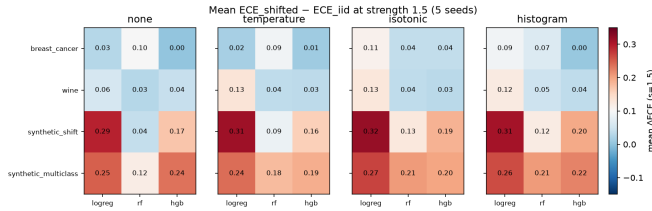


Figure 2: Mean ΔECE at $s=1.5$. Positive (red) means the shifted test is *worse* calibrated than the i.i.d. test.

5.2 Post-hoc maps are not a reliable remedy

Figure 3 plots $ECE_{\text{shift(cal)}} - ECE_{\text{shift(none)}}$. Wine random forests *benefit* from isotonic (-0.123 ECE; shifted ECE 0.175 $\rightarrow$ 0.052), because the uncalibrated forest was already badly calibrated. Temperature scaling on the same cell reaches shifted ECE 0.068. Breast-cancer logistic regression is *hurt* by isotonic (0.074). We therefore do not rank calibrators under shift.

5.3 Coverage under perturbation

Unweighted split conformal (logreg, none, LAC score $1 - p_y$) coverage at $\alpha = 0.1$ (seed mean $\pm$ sd): breast-cancer 0.913 ± 0.015 ($s=0$), 0.778 ± 0.045 ($s=1.5$), 0.608 ($s=2.5$); synthetic binary 0.634 ± 0.190 at $s=1.5$ (Figure 4). Exchangeability is broken by construction. A 1-D “oracle” density ratio on feature 0 did *not* restore coverage in the H3 pilot (killed).

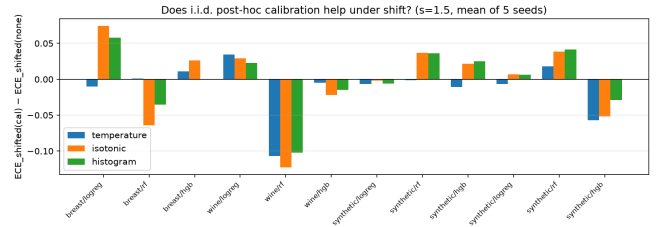


Figure 3: Does i.i.d. calibration help on the shifted test? Negative bars mean the calibrator beats *none* under shift.

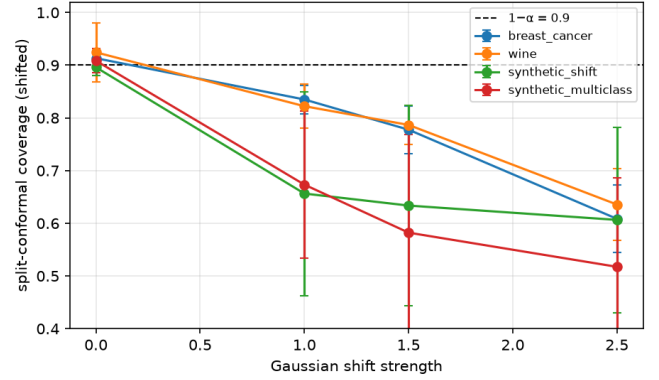


Figure 4: Unweighted split-conformal coverage vs. feature perturbation (logreg, no calibrator, LAC score). Dashed line: $1 - \alpha = 0.9$.

5.4 Selection shift is a different, weaker effect

Quantile slicing and importance resampling keep original (X, y) pairs, so $P(Y | X)$ is preserved and the mechanism is closer to covariate shift. This ablation uses 6 dataset $\times$ model cells on 2 datasets (breast-cancer and synthetic_shift; wine and the multiclass synthetic are absent). Mean ΔECE (none) is 0.016 (sd 0.006) for quantile slice and 0.011 (sd 0.031) for importance resampling. On the *same* 6 cells, Gaussian perturbation at $s=1.5$ has cell-mean ΔECE 0.105 (sd 0.109). Quantile slice also moves accuracy less (cell-mean Δacc -0.016 vs. matched Gaussian -0.128), so the gap is not identified as “preserving $P(Y | X)$ ” alone. Importance resampling can even *decrease* ECE on breast-cancer. This control **does not support** a claim that i.i.d. post-hoc calibration fails under covariate shift in general. Trailing-column perturbation on synthetic_shift yields RF $\Delta ECE = 0.009$ vs. 0.041 when perturbing the first four columns.

6 Limitations

- **The headline mechanism is not covariate shift.** Frozen-label feature mean-shift changes $P(Y | X)$ at the observed x (input corruption). Selection controls that preserve pairings show small ΔECE . Title and abstract are scoped accordingly.
- **Shift is synthetic** and, for feature perturbation, entan-

gled with accuracy drop (synthetic logreg 0.932→0.620 at $s=1.5$).

- **Small tabular data only.** Wine test splits have 45 points and 44 calibration points with 15 ECE bins; those cells are noisy. No ImageNet, no LMs, no naturally occurring drift (no Folktables/WILDS).
- **Multiclass isotonic/histogram** map confidence, not a full Dirichlet map [16]; argmax can flip after renormalization.
- **Temperature on probabilities.** Trees have no logits; RF 0/1 probabilities saturate after clipping. This is not Guo logit temperature.
- **H3 weights were misspecified.** A 1-D histogram ratio is not $p_{\text{test}}(x)/p_{\text{cal}}(x)$ for a 4-D perturbation.
- **Pooling and tests.** Seed-level $n = 60$ Wilcoxon is pseudo-replicated and one-sided. The $n = 12$ seed-averaged uncalibrated means are a sign count, not independent replicates. A dataset-level one-sided sign test has $n = 4$ and $p=0.0625$. 22 tests, unadjusted. Selection comparisons use 6 cells on 2 datasets and are not strength-matched to $s=1.5$ perturbation.
- **Planar peg-in-hole cartoon, not a robot.** Dual-regime numbers are from `exp09_peg_insert` (HistGradient-Boosting, 5 seeds). Geometric clearance only: no MuJoCo, no DexNet, no delayed-label receding calibrator in the homogeneous grid. Pre-registered kill criteria fired: none.
- **Search scope.** Novelty is relative to `paper/verified.bib`.

7 Dual-regime insert/abort

The tabular measurement says i.i.d. maps fail under frozen-label perturbation, not under pairing-preserving selection. A hiring-relevant task is whether that split changes *what the agent does*, with asymmetric cost: a false-confident insert is much worse than deferring or aborting ($u_{\text{wrong}} = -10$, $u_{\text{correct}}=1$, $u_{\text{defer}} = -0.2$, $u_{\text{abort}} = -0.5$).

We use a planar peg-in-hole cartoon (`src/dualregime/`). True success is geometric clearance: $\text{hypot}(x, y) + k|\theta| \leq r_{\text{clear}}$. The deployed model sees encoder pose plus an unused appearance channel. Camera pose is a watchdog, not a training feature. *Optimistic encoder bias* scales encoder (x, y) toward the origin and freezes Y ; near-origin poses exist in training, so a PCA residual on encoder coordinates *falls* and is the wrong detection channel. *Selection* keeps pairs with $x \geq 0$ (right-half fixture). Success rates are difficulty-matched (0.458 i.i.d. vs. 0.426 slice).

Two channels, opposite legal moves. Physics residual (encoder-camera, plus $\|p(\text{raw}) - p(\text{projected})\|_1$) routes to **project** the encoder onto the camera pose, then apply source temperature and unweighted LAC: act if the set is {success},

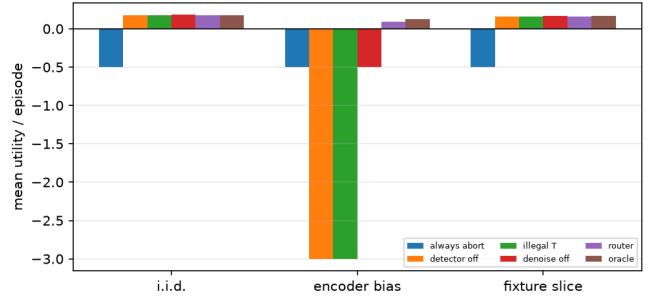


Figure 5: Mean insert/abort utility by test split and policy (5 seeds). Optimistic encoder bias is not a fixture slice.

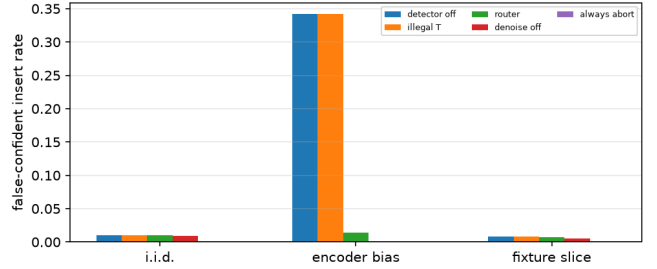


Figure 6: False-confident insert rate. Source T on raw encoder probabilities is the incident; projecting then applying T is not abort-all.

else defer; abort only if residual is huge. Density ratio on camera xy (a domain classifier, not the killed 1-D feature-0 H3 estimator) routes to the *same* source T plus multivariate weighted LAC: act or **defer**, never abort-as-sensor-fault, never project as if the encoder lied. Applying source T to raw corrupted encoder probabilities is an ablation (`illegal_T / detector_off`), not the perturbation policy.

Figure 5–6 and `results/summary_dual.json` (5 seeds, HGB). Under encoder bias, router mean utility is 0.093 vs. illegal raw- T -3.000 vs. abort-all -0.500 vs. denoise-off (abort, no project) -0.500. FCAR drops from 0.342 to 0.014; insert rate remains 0.396 (abort rate 0.110, the huge-residual tail). Oracle/always-project utility is 0.126: the small gap is that tail, not a second camera classifier. On i.i.d. tests the router does not pay a large tax (0.177 vs. 0.176). On the fixture slice, abort rate is 0.000 and utility 0.158 beats abort-all (-0.500). Weighted conformal does not have to *help* here; it is enough that we do not treat selection as a broken encoder. Mixed-stream blocks (Figure 7) keep those moves: slice-block density label rate 0.949, bias-block residual label rate 0.929, project-then- T rate 0.804. Kill criteria fired: none. This is a kinematic cartoon, not hardware [22].

8 Conclusion

I.i.d. post-hoc maps are not a dependable answer to frozen-label feature perturbation on these sklearn models, and they are not a dependable *insert* policy when the deployed encoder

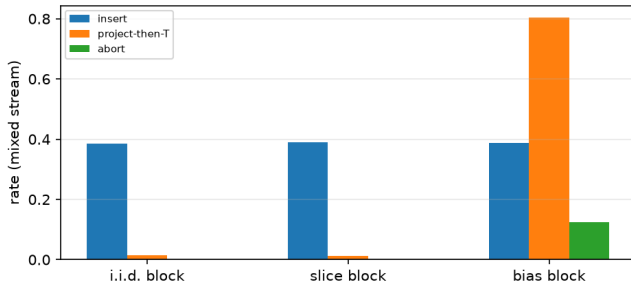


Figure 7: Mixed stream (i.i.d. then slice then encoder bias). Rolling window; not one batch over the concatenated stream.

is the sensor that moved. A residual channel that projects onto the camera pose recovers utility under that perturbation and is not equivalent to abort-all or to temperature scaling on raw corrupted probabilities. Pairing-preserving selection is a different mechanism and a different legal move (defer, not abort). The HGB exception and the killed H2/H3 hypotheses remain part of the tabular measurement.

AI-generation disclosure

This repository and draft were produced by a coding agent executing the user’s GOAL.md protocol (literature verification via arXiv and Semantic Scholar APIs, experiments on CPU, no hand-edited results/*.json). Every numeric macro in paper/numbers.tex is generated from those files. Citations are restricted to paper/verified.bib. A human should still read the limitations and re-run make fresh-clone-test before treating the PDF as camera-ready.

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